

Exploratory Data Analysis

Create a defensible path from messy observations to analytical evidence.



Learning outcomes

01  Separate descriptive patterns from causal claims.

02  Choose comparisons that fit the question.

03  Document anomalies and competing explanations.

The concept system

01

distributions

02

group comparisons

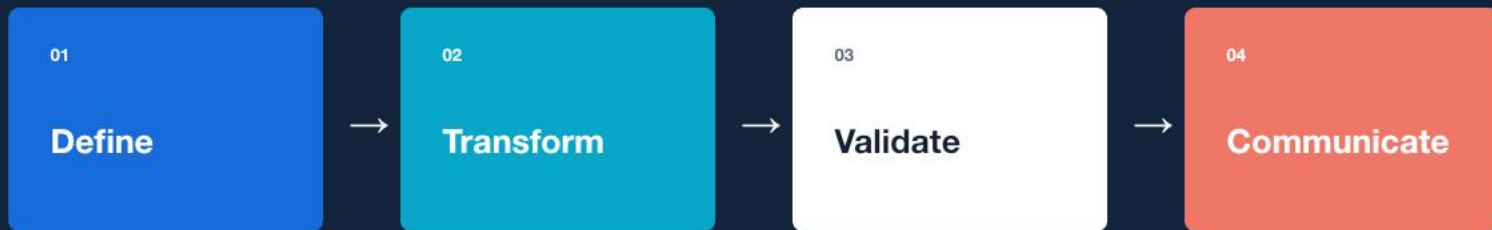
03

correlation

04

bias

A repeatable reasoning loop



Investigate retention differences and produce an evidence notebook.

Guided lab

LAB BRIEF

Investigate retention differences and produce an evidence notebook.

workbook / tests / evidence

01

Frame

Define inputs, outputs, and success criteria.

02

Build

Implement the smallest correct workflow.

03

Verify

Test normal, boundary, and failure cases.

04

Explain

Record the evidence and remaining limits.

Portfolio deliverable

Five findings, each paired with evidence and a limitation.

QUALITY GATE

- **Reproducible**
- **Tested**
- **Decision-relevant**

WEEKLY ASSESSMENT

Evidence-First EDA

Cleaning 30%, reasoning 30%,
visual evidence 20%, limitations
10%, reproducibility 10%.

What should remain after the lesson?

CONCEPT

distributions + group comparisons

PRACTICE

Investigate retention differences and produce an evidence notebook.

EVIDENCE

Five findings, each paired with evidence and a limitation.
